

AYUSH NAVJEEVAN DHOTE

B.Tech Bioengineering | MIT ADT University, Pune | CGPA: 7.97

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PROFILE

Final-year Bioengineering student at MIT ADT University with hands-on experience in Python-based bioinformatics pipelines, machine learning, molecular docking, and biomedical device design. Passionate about bridging computation and biology — currently building toward an AI-driven drug repurposing pipeline targeting neglected tropical diseases. Active GitHub contributor, NPTEL IIT Madras Bioinformatics certified, and experienced across both wet-lab and computational research environments.

EDUCATION

B.Tech in Bioengineering — School of Bioengineering Sciences & Research, MIT ADT University, Pune CGPA: 7.97 | 2023–2027

SKILLS & DOMAINS

Programming	Python (Advanced) · SQL · R · NoSQL (MongoDB)
ML / AI	Machine Learning · Data Mining · Bioinformatics Pipelines · NCBI Entrez API
Domains	Computational Biology · Drug Discovery · Biomedical Device Design · Molecular Docking · Microfluidics
Instrumentation	HPLC · TLC · GC · PCR/qPCR · Electrophoresis · Microscopy
Wet Lab	DNA/RNA Extraction · Cell Culture · Chromatography · Enzyme Assays
Embedded	Arduino (C/C++) · Microcontroller Programming (8051, AVR) · Electronics Fundamentals
Tools & Infra	MongoDB · PyMOL · AutoDock Vina · PowerBI · Cytoscape · Ubuntu/Linux
Fluent	English · Hindi · Marathi
Learning	German · Japanese

CERTIFICATIONS & WORKSHOPS

- NPTEL IIT Madras — Bioinformatics (Certified)
- Smart India Hackathon — Multiple participations (2024, 2025)
- MEDHA Medical Device Hackathon — IIT Bombay
- ICRTB Workshops — 2024, 2025, 2026

PROJECTS

NCBI Gene Data Pipeline *[Python · Biopython · NCBI Entrez API · CSV · MongoDB]*

- Built an automated pipeline to fetch and process Alzheimer's disease-related gene records directly from NCBI using Biopython's Entrez API — replacing manual database browsing with a reproducible, scriptable workflow.

- Parses raw XML/GenBank response data and structures clean gene metadata — including gene IDs, full descriptions, chromosomal locations, and aliases — into analysis-ready CSV files.
- Currently extending the project with Rosalind bioinformatics algorithm challenges to deepen sequence analysis and data parsing skills.
- Published on GitHub (AYUSHBEINGCREATIVE/ncbi-gene-pipeline). Serves as the first stage of a planned full AI drug repurposing pipeline: gene retrieval → PDB structure fetch → AutoDock Vina docking → MongoDB storage → analysis.

Rosalind Bioinformatics Problem Set [\[Python · Algorithms · Sequence Analysis \[ACTIVE\]\]](#)

- Actively solving Rosalind bioinformatics problems from scratch — currently working through problems covering DNA string operations, GC content, Hamming distance, RNA transcription, and complementary strand generation.
- Every solution is independently written and each problem builds a concrete skill in biological sequence manipulation and algorithm design.
- Progress published and continuously updated on GitHub (AYUSHBEINGCREATIVE/rosalind-solutions), demonstrating consistent learning momentum.

Real-Time Data Dashboard [\[Python · MongoDB · Plotly Dash · Ubuntu · Power BI\]](#)

- Built a live end-to-end data pipeline connecting MongoDB (hosted on WSL/Ubuntu) to a Plotly Dash web dashboard — enabling real-time data visualisation without manual data exports.
- Dashboard includes KPI summary cards, six interactive chart types, and a fully filterable data table that refreshes automatically as new records enter the database.
- Integrated MongoDB BI Connector to also expose the same data to Power BI, allowing dual reporting from a single data source.

Non-Invasive Blood Group Detection Device [\[SIH Ideathon 2024 · Optical Biosensing · Biomedical Engineering\]](#)

- Designed a non-invasive blood group identification system capable of detecting blood type without any physical blood collection — eliminating needles, reagents, and contamination risk.
- Core innovation: a high-resolution optical sensor precisely tuned to the spectral fingerprints of surface antigens — specifically the A, B, and H antigens expressed on the outer membrane of red blood cells. These antigens interact differently with structured light depending on their molecular composition and density.
- The system works by directing a controlled, narrow-spectrum light source through the skin over a superficial capillary bed. As light interacts with the antigen structures on RBC surfaces, it produces unique absorption and scattering patterns. The sensor captures these patterns with sufficient resolution to discriminate between antigen types — enabling algorithmic classification of blood group (A, B, AB, or O) in real time.
- The key design challenge addressed was achieving sensor sensitivity high enough to resolve antigen-level differences at the cellular scale — bridging optical physics with haematology in a wearable-compatible format.
- Presented at Smart India Hackathon Ideathon 2024.

Temperature-Controlled Bone Drilling System [\[MEDHA · IIT Bombay · Orthopaedic Device · Nanotechnology\]](#)

- Designed an active-cooling orthopaedic drilling system engineered specifically to prevent thermal necrosis — the irreversible death of bone cells (osteocytes) that occurs when drilling temperature exceeds approximately 47°C, a well-documented threshold in orthopaedic surgery literature.
- The drill incorporates a co-axial nano-coolant delivery channel running alongside the drill bit shaft, enabling continuous localised heat dissipation directly at the bone-cutting interface — where thermal damage is most concentrated.
- Used TiO₂ (titanium dioxide) nanoparticles as the active cooling agent in the coolant fluid — selected for their documented biocompatibility in orthopaedic implant environments, high thermal conductivity at nanoscale, and chemical stability. TiO₂ nanofluids are supported in drilling and biomedical literature for their ability to enhance heat removal without adverse tissue interaction.
- A closed-loop temperature monitoring circuit tracks real-time drill-site temperature and modulates coolant flow dynamically — maintaining bone temperature within safe thresholds throughout the procedure and reducing risk of delayed healing or implant failure.
- Presented at MEDHA Medical Device Hackathon, IIT Bombay.

Medicine Supply Chain Optimization Platform [\[SIH 2024 · Software · Healthcare Logistics\]](#)

- Designed a digital traceability and inventory management platform to address two critical failures in pharmaceutical distribution: counterfeit drug infiltration and unpredictable stockouts that leave patients without essential medicines.
- Each medicine batch is assigned a unique identifier tracked across every node in the supply chain — manufacturer, regional warehouse, distributor, and pharmacy — giving regulatory and logistics stakeholders complete end-to-end visibility.
- The platform incorporates predictive inventory alerts based on consumption patterns, flagging potential shortfalls before they occur, along with automated expiry tracking to eliminate wastage of near-expiry stock.
- Cold-chain integrity monitoring ensures temperature-sensitive medicines (vaccines, biologics) remain within required storage ranges throughout transit, with alerts triggered on deviation.
- Participated in Smart India Hackathon 2024.

Bamboo–Plastic Composite Road Panels *[SIH 2025 · Materials Engineering · NER Infrastructure [WINNER — ICRTB Expo 2026]]*

- Developed modular road panels fabricated from a bamboo–plastic composite material, purpose-built for the North East Region (NER) of India — a geography defined by high annual rainfall, seasonal flooding, and terrain that makes conventional road construction costly and frequently damaged.
- The composite leverages bamboo's exceptional tensile strength and natural abundance in NER alongside plastic's high water resistance and durability — producing a panel that outperforms conventional materials in wet, high-stress conditions without relying on imported or scarce resources.
- Critically designed for flood resilience and field repairability: individual panels are fully modular and interchangeable. When a section is flood-damaged or structurally compromised — a routine occurrence in NER districts — only that specific panel is replaced, eliminating the need for full road reconstruction.
- The assembly system is designed for single-point repair: even a partially broken panel can be detached, swapped, or reassembled on-site without heavy machinery or specialised crews, drastically reducing maintenance downtime and logistics cost in remote areas.
- Winner — ICRTB Expo 2026.

VOLUNTEER EXPERIENCE

- Volunteer — ICRTB Conference, MIT ADT University, Pune
- Volunteer — SBSR Induction Program, MIT ADT University
- Volunteer — International Women's Day Event

ACHIEVEMENTS & EXTRA-CURRICULAR

- **First Runner-Up (Group) — Persona Showstopper 2026:** Anchored the performance as a lead male model, collaborating on group choreography, runway presentation, and creative execution to secure a podium finish.
- **Winner — ICRTB Expo 2026:** Awarded First Place for an innovative Bamboo–Plastic Composite Road Panels engineering project.
- Two National-Level Silver Medals in Skating
- Listed in Genius / Asia / Limca Book of Records for Skating achievements
- Active in music, dance, and cultural performances